

# Pregame Pitcher Strikeout Projection — Project Summary

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Companion to the full technical manuscript in `docs/paper/manuscript.md`.

## Goal

Estimate a starter’s strikeout rate before first pitch, project how many batters that starter will face, and convert the pair into expected strikeouts and  $P(K \geq L)$  for common prop lines  $L$ —using only information available before first pitch:

$$k_{\text{rate}} \times \text{TBF} \rightarrow E[K] \rightarrow P(K \geq L)$$

## Models and headline metrics

| Component           | Model                               | Role                         |
|---------------------|-------------------------------------|------------------------------|
| Strikeout rate      | Unweighted LightGBM (180 features)  | Frozen production rate model |
| Batters faced (TBF) | Ridge (thin bullpen, 24 features)   | Projected exposure           |
| Counts / lines      | Binomial / Poisson on projected TBF | Expected K and $P(K \geq L)$ |

**Frozen rate model** (2023–2024 fit; chronological test from 2024-08-06): MAE / RMSE /  $R^2 \approx$  **0.0787 / 0.0987 / 0.147** (~15% of game-level rate variance). Same test: train-mean MAE  $\approx$  **0.0854**; Marcel-lite talent baseline MAE  $\approx$  **0.0826**.

**Count layer** (same test partition, projected-TBF exposure): expected-K MAE / RMSE /  $R^2 \approx$  **1.790 / 2.213 / 0.168**; line Briers roughly **0.12–0.22** (lines 3.5–7.5).

## Full-stack evaluation

| Evaluation gate             | Result                                                                                                             |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------|
| Estimator tuning            | Keep baseline LightGBM defaults; Ridge $\alpha$ tuned and persisted                                                |
| Walk-forward stack backtest | Mean expected-K MAE $\approx$ <b>1.778</b> (3 expanding 2024 windows; $\sigma \approx 0.036$ )                     |
| Calibration                 | Mean ECE $\approx$ <b>0.024</b> (internal chronological calibration; no sportsbook or public-projection benchmark) |
| Population policy           | Metrics conditional on $PA \geq 9$ (~3.5% of first pitchers excluded)                                              |

## Takeaway

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Built and validated a leakage-safe Statcast → features → rate × exposure stack with nested chronological selection and automated leakage tests. Predictive power is modest (rate  $R^2 \approx 0.15$ ) but clears a Marcel-lite season-talent floor on the same chronological test. Leave-family-out results are reported per outer fold: lineup is both-fold positive for both models; several other mean deltas flip sign. The portfolio claim is ML engineering discipline under a hard pregame constraint—not a strong or market-beating predictor. Next: pregame role labels, post-freeze holdout, closing-line evaluation if practical value is the goal.

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*Full write-up: [docs/paper/manuscript.pdf](#)*